

STATISTIC

CHAPTER 4

Exercise 4.1

$$f(n) = \begin{cases} kn, & 0 \leq n \leq 4 \\ 0, & \text{elsewhere} \end{cases}$$

a. Determine constant k

$$\int_0^4 kn = 1$$

$$k \left[\frac{n^2}{2} \right]_0^4 = 1$$

$$k \left[\frac{4^2}{2} - 0 \right] = 1$$

$$k(8) = 1$$

$$8k = 1$$

$$k = \frac{1}{8} \neq$$

c. what is the probability that students will see more than 2 hours?

$$\int_2^4 n/8$$

$$= \left[\frac{n^2}{8(2)} \right]_2^4$$

$$= \left[\frac{4^2}{16} - \frac{2^2}{16} \right] = \frac{3}{4} \neq$$

Exercise 4.2

$$f(n) = \begin{cases} c(n-1)(2-n), & 1 \leq n \leq 2 \\ 0, & \text{elsewhere} \end{cases}$$

b. what is the probability that student will see their academic between 1 and 2 hours?

a. Determine the constant c.

$$(n-1)(2-n)$$

$$= 2n - n^2 - 2 + n$$

$$= -n^2 + 3n - 2$$

$$\int_1^2 \frac{n}{8}$$

$$= \left[\frac{n^2}{8(2)} \right]_1^2$$

$$= \left[\frac{2^2}{16} - \frac{1^2}{16} \right]$$

$$= \frac{3}{16} \neq$$

$$c \int_1^2 -n^2 + 3n - 2 = 1$$

$$c \left[-\frac{n^3}{3} + \frac{3n^2}{2} - 2n \right]_1^2 = 1$$

$$c \left[\left(-\frac{2^3}{3} + \frac{3(2)^2}{2} - 2(2) \right) - \left(-\frac{1^3}{3} + \frac{3(1)^2}{2} - 2(1) \right) \right] = 1$$

$$c \left[\frac{1}{6} \right] = 1$$

$$c = 6 \neq$$

b. Determine the probability that a student will finish the test in less than 65 minutes?

$$\int_1^{1.0833} 6(-n^2 + 3n - 2)$$

$$= 6 \left[\frac{-n^3}{3} + \frac{3n^2}{2} - 2n \right]_1^{1.0833}$$

$$= 6 \left[\frac{-(1.0833)^3}{3} + \frac{3(1.0833)^2}{2} - 2(1.0833) - \left(\frac{-1^3}{3} + \frac{3(1)^2}{2} - 2(1) \right) \right]$$

$$= 6[-0.830059 - 13/6]$$

$$= 0.0197$$

c. Determine the probability that a student will finish the test between 1.2 to 1.6 hours

$$\int_{1.2}^{1.6} 6(-n^2 + 3n - 2)$$

$$= (-2(1.6)^3 + 9(1.6)^2 - 12(1.6)) -$$

$$(-2(1.2)^3 + 9(1.2)^2 - 12(1.2)) -$$

$$4.352 - (-4.896)$$

$$= 0.544$$

Exercise 4-3

$$f(n) = \begin{cases} 3k\sqrt{n}, & 0 < n < 1 \\ 0, & \text{elsewhere} \end{cases}$$

a. Evaluate k

$$k \int_0^1 3n^{1/2} = 1$$

$$3k \left[\frac{(2)n^{3/2}}{3} \right]_0^1 = 1$$

$$3k \left(\frac{2(1)^{3/2}}{3} - 0 \right) = 1$$

$$3k \left(\frac{2}{3} \right) = 1$$

$$k = \frac{3}{2}$$

b. Find the cumulative distribution function and use it to evaluate $P(0.4 < n < 0.7)$

$$f(n) = \frac{3}{2} \int_{0.4}^{0.7} n^{1/2} dn$$

$$f(0.4) = \frac{3}{2} \left[\frac{(2)n^{3/2}}{3} \right]_{0.4}^{0.7}$$

$$= \frac{3}{2} \left(\frac{2(0.7)^{3/2}}{3} - \frac{2(0.4)^{3/2}}{3} \right)$$

$$= 0.25298$$

$$= 0.58566 - 0.25298$$

$$= 0.3327$$

Exercise 4.4

$$f(n) = \begin{cases} 0, & n < 0 \\ 0.2n, & 0 \leq n < 5 \\ 1, & n \geq 5 \end{cases}$$

a. $P(n < 3.4)$

$$\begin{aligned} f(3.4) &= 0.2n \\ &= 0.2(3.4) \\ &= 0.68 \end{aligned}$$

b. $P(0.3 < n < 4.2)$

$$\begin{aligned} f(n) &= 0.2(4.2) - 0.2(0.3) \\ &= 0.84 - 0.06 \\ &= 0.78 \end{aligned}$$

c. $P(n > 2.4)$

$$\begin{aligned} f(n) &= 1 - 0.2(2.4) \\ &= 1 - 0.48 \\ &= 0.52 \end{aligned}$$

Exercise 4.5

$$f(n) = \begin{cases} 3/n^4, & n \geq 1 \\ 0, & \text{otherwise} \end{cases}$$

Find mean and variance of n

$$\int_1^{\infty} \frac{3}{n^4} dn = 1$$

mean $(E[n]) =$

$$\begin{aligned} \int_1^{\infty} n \cdot \frac{3}{n^4} dn &= 3 \int_1^{\infty} n^{-3} dn \\ &= 3 \left[\frac{n^{-2}}{-2} \right]_1^{\infty} \end{aligned}$$

$$\begin{aligned} &= 3 \left[-\frac{1}{2} \right] \\ &= 3 \left(0 - \left(-\frac{1}{2} \right) \right) \\ &= \frac{3}{2} \end{aligned}$$

variance $(E[n^2]) =$

$$\begin{aligned} \int_1^{\infty} n^2 \cdot \frac{3}{n^4} dn &= 3 \int_1^{\infty} n^{-2} dn \\ &= 3 \left[-\frac{n^{-1}}{1} \right]_1^{\infty} \end{aligned}$$

$$\begin{aligned} &= 3 \left(0 - (-1) \right) \\ &= 3 \end{aligned}$$

variance

$$\begin{aligned} &= E[n^2] - (E[n])^2 \\ &= 3 - \left(\frac{3}{2} \right)^2 \end{aligned}$$

$$= 3 - \frac{9}{4}$$

$$= \frac{3}{4}$$

Exercise 4-6

$$f(y) = \begin{cases} \frac{3}{32} (y-1)(5-y), & 1 \leq y \leq 5 \\ 0, & \text{otherwise} \end{cases}$$

Find expected value and
standard deviation of y

Expected value ; $E(y)$:

$$\begin{aligned} & \int_1^5 y \cdot f(y) dy \\ &= \int_1^5 y \cdot \left[\frac{3}{32} (y^2 + 6y - 5) \right] dy \\ &= \int_1^5 y \left(\frac{-3y^2}{32} + \frac{18y}{32} - \frac{15}{32} \right) dy \\ &= \int_1^5 \left(\frac{-3y^3}{32} + \frac{18y^2}{32} - \frac{15y}{32} \right) dy \\ &= \int_1^5 \left(\frac{-3y^3}{32} + \frac{18y^2}{32} - \frac{15y}{32} \right) dy \end{aligned}$$

$$\begin{aligned} &= \left[\frac{-3y^4}{32(4)} + \frac{18y^3}{32(3)} - \frac{15y^2}{32(2)} \right]_1^5 \\ &= \left[\frac{-3(5)^4}{128} + \frac{6(5)^3}{32} - \frac{15(5)^2}{64} \right] - \left[\frac{-3(1)^4}{128} + \frac{6(1)^3}{32} - \frac{15(1)^2}{64} \right] \\ &= \left[\left(\frac{375}{128} \right) - \left(\frac{9}{128} \right) \right] \\ &= 3 \end{aligned}$$

Standard deviation

$$E(y) = 3$$

$$E(y^2) = \int_1^5 y^2 \left(\frac{-3y^2}{32} + \frac{18y}{32} - \frac{15}{32} \right) dy$$

$$= \int_1^5 \left(\frac{-3y^4}{32} + \frac{18y^3}{32} - \frac{15y^2}{32} \right) dy$$

$$= \left[\frac{-3y^5}{32(5)} + \frac{18y^4}{32(4)} - \frac{15y^3}{32(3)} \right]_1^5$$

$$= \left(\frac{-3(5)^5}{160} + \frac{18(5)^4}{128} - \frac{5(5)^3}{32} \right) -$$

$$\left(\frac{-3(1)^5}{160} + \frac{18(1)^4}{128} - \frac{5(1)^3}{32} \right)$$

$$= \frac{625}{64} - \left(-\frac{11}{320} \right) = \frac{49}{5}$$

$$\text{variance} = E(y^2) - (E(y))^2$$

$$= \frac{49}{5} - 3^2$$

$$= 0.8$$

Standard

deviation, $\sigma = \sqrt{0.8}$

$$= 0.8944$$

Exercise 4-7

Find the probabilities for the standard normal z

a. $P(-1.35 < z < 2.15)$

$$P(z < 2.15) - P(z < -1.35)$$

$$= 0.9842 - 0.0885$$

$$= 0.8957 \quad \#$$

b. $P(0 < z < 2.37)$

$$P(z < 2.37) - P(z < 0)$$

$$= 0.9911 - 0.5000$$

$$= 0.4911 \quad \#$$

c. $P(z > 0.96)$

$$1 - P(z < 0.96)$$

$$= 1 - 0.8315$$

$$= 0.1685 \quad \#$$

d. $P(z < -5.20)$

$$= 1.0000 \quad \#$$

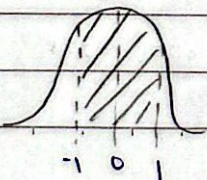
e. $P(-0.5 < z < 2.3)$

$$P(z < 2.3) - P(z < -0.5)$$

$$= 0.9893 - 0.3085$$

$$= 0.6808 \quad \#$$

f. $P(z < -4.22) = 0 \quad \#$

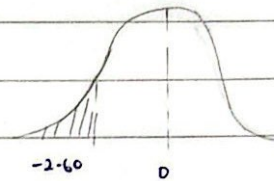
Exercise 4-8

Assume z has a standard normal distribution

Sketch and determine value z .

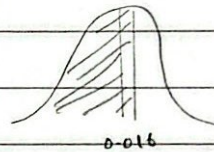
a. $P(z < z) = 0.0047$

$$z = -2.60$$



b. $P(z < z) = 0.5636$

$$z = 0.16$$



c. $P(-2 < z < z) = 0.93$

$$P(-2) = (1 - 0.93) / 2$$

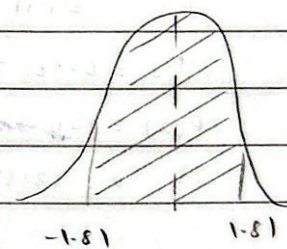
$$P(-2) = 0.035$$

$$-z = -1.81$$

$$P(z) = 0.93 + 0.035$$

$$P(z) = 0.965$$

$$z = 1.81 \quad \#$$



d. $P(-2 < z < z) = 0.6826$

$$P(-2) = (1 - 0.6826) / 2$$

$$P(-2) = 0.1587$$

$$-z = -1.0$$

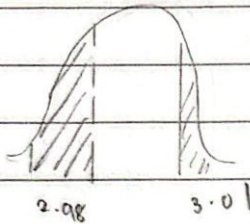
$$P(z) = 0.6826 + 0.1587$$

$$P(z) = 0.8413$$

$$z = 1.0$$

Exercise 4.9

A machine produce tops whose diameters are normally distributed. Diameter either shorter than 2.98 or longer than 3.01 what percentage of all top



$$Z = \frac{X - \mu}{\sigma}$$

$$(Z < 2.98) = \frac{2.98 - 3}{0.009}$$

$$Z = -2.222$$

$$P(Z) = 0.0132$$

$$(Z < 3.01) = \frac{3.01 - 3}{0.009}$$

$$= 1.11$$

$$P(Z) = 0.3665$$

$$P(Z) = 1 - 0.8665$$

$$= 0.1335$$

$$= 0.0132 + 0.1335$$

$$= 0.1467$$

$$= 14.67\%$$

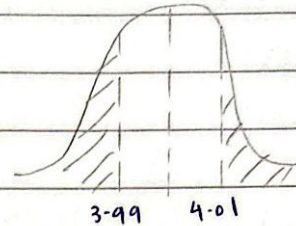
Exercise 4.10

$$\text{length} = 4.0 \pm 0.01 \text{ cm}$$

$$\text{mean} = 4.0$$

$$\text{std.} = 0.005$$

5000 nails, how nails will rejected



$$\mu = 4$$

$$\sigma = 0.005$$

$$\text{range} = 4.0 + 0.01 = 4.01$$

$$4.0 - 0.01 = 3.99$$

$$(Z < 3.99) = \frac{3.99 - 4}{0.005}$$

$$= -2.0$$

$$P(Z) = 0.0228$$

$$(Z > 4.01) = \frac{4.01 - 4}{0.005}$$

$$= 2$$

$$P(Z) = 0.9772$$

$$= 1 - 0.9772$$

$$= 0.0228$$

$$P(\text{rejection}) = 0.0228 + 0.0228$$

$$= 0.0456$$

$$\text{Number} = 5000 \times 0.0456 = 228$$

rejected